PortfolioGenie – Project Proposal

Graduation Project | Digital Egypt Pioneers Initiative (DEPI)
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1. Project Overview

PortfolioGenie is a web application designed to help junior developers create professional, personalized portfolios with ease. The platform aims to simplify the process of presenting technical skills, projects, and professional experience in a clear and attractive way.

By integrating GitHub data and AI-powered content generation, PortfolioGenie aims to help users transform their existing projects and coding activities into meaningful portfolio content. Users will be able to review, edit, and customize their information before presenting their portfolios.

The project will be developed as part of the Digital Egypt Pioneers Initiative (DEPI) graduation project, allowing the team to apply its web development knowledge and software engineering practices to a practical application.

2. Problem Statement

Junior developers often struggle to build professional portfolios that effectively showcase their skills, projects, and technical experience. Creating a portfolio from scratch can require web development, design, and content-writing skills, as well as considerable time and effort.

In addition, developers may have valuable projects and public coding activity on GitHub but find it difficult to turn that information into clear project descriptions and professional personal profiles.

PortfolioGenie aims to address these challenges by providing a beginner-friendly platform that simplifies portfolio creation, supports customization, and assists users in presenting their technical work more effectively.

3. Proposed Solution

PortfolioGenie will provide a web-based platform that guides users through creating and customizing a professional portfolio without requiring them to build a website from scratch.

The proposed solution will focus on the following capabilities:

- Guided Portfolio Creation: A step-by-step process for entering personal information, skills, and project details.
- GitHub Integration: Retrieving selected publicly available profile and repository information to help users showcase their work.
- AI-Assisted Content Generation: Generating suggested professional biographies and project descriptions based on available user information.
- Live Preview and Customization: Allowing users to preview their portfolios and customize their content and appearance.
- User Control: Enabling users to review and edit generated content before using it in their portfolios.

The final features and technical implementation will be confirmed during the requirements and planning phases.

4. Project Objectives

- Develop a functional and user-friendly web application for portfolio creation.
- Simplify the process of presenting technical skills, projects, and professional information.
- Integrate GitHub data to help users showcase their publicly available development work.
- Explore AI-assisted content generation to improve the clarity and presentation of portfolio content.
- Provide portfolio customization and a responsive interface across different screen sizes.
- Apply software development practices, including requirements analysis, implementation, testing, documentation, and teamwork.
- Deliver a tested application and demonstrate its core functionality by the end of the graduation project.

5. Target Users

The primary target users are junior developers and individuals who want to present their technical work professionally.

Potential user groups include:

- Computer science and IT students.
- Fresh graduates seeking their first development opportunities.
- Junior developers building their professional presence.
- Freelancers who need to showcase their projects and technical skills.

6. Project Scope

In Scope – Proposed Minimum Viable Product (MVP)

- A web application for creating and customizing professional portfolios.
- A guided form for entering personal information, skills, and project details.
- GitHub integration for retrieving selected public profile and repository information.
- AI-assisted generation of professional biographies and project descriptions.
- A live portfolio preview and responsive presentation.
- Testing, technical documentation, and a final project demonstration.

Out of Scope or Subject to Approval

- Advanced portfolio themes and extensive design customization.
- Automated analysis of private GitHub repositories.
- Paid subscription plans or payment processing.
- Advanced analytics and portfolio performance tracking.
- Any additional features not approved during requirements gathering.

The team will prioritize the core features according to the available time and resources. The final scope will be agreed upon with the instructor before implementation.

7. Expected Outcomes

- A working web application that supports the agreed core portfolio-building features.
- A user-friendly process for creating and customizing portfolio content.
- GitHub integration and AI-assisted content generation, if confirmed within the approved scope.
- A responsive application tested against its main functional and usability requirements.
- Project documentation covering requirements, design, implementation, testing, and setup instructions.
- A final presentation demonstrating the application and each team member's contribution.

8. Proposed Technology Stack

The following technologies are proposed for developing PortfolioGenie. The final technology stack will be confirmed during the planning and requirements phases.

- **Front-End:** React.js

- **Styling:** Tailwind CSS
- **Back-End:** PHP and Laravel
- **Database:** MySQL
- **GitHub Integration:** GitHub REST API
- **AI Content Generation:** An AI API for generating professional biographies and project descriptions.
- **API Communication:** RESTful APIs to connect the front-end with the back-end.
- **Version Control:** Git and GitHub
- Testing: Appropriate tools for testing functionality, API integration, and responsiveness.

The final implementation details will be confirmed during the requirements and planning phases based on the project scope and team capabilities.

9. Items to Be Confirmed by the Team

- The final portfolio sections and customization options.
- Whether user accounts and persistent data storage are required.
- How portfolios will be saved, published, or shared.
- The GitHub data to be retrieved and the integration method.
- The AI service, its limitations, and how generated content will be reviewed.
- The selected front-end, back-end, and database technologies.
- The project timeline, milestones, and responsibilities of each team member.